

Stable Diffusion

Naeemullah Khan

naeemullah.khan@kaust.edu.sa



جامعة الملك عبد الله
للعلوم والتقنية

King Abdullah University of
Science and Technology

KAUST Academy
King Abdullah University of Science and Technology

July 3, 2026

1. Stable Diffusion overview: text-to-image foundations and key components
2. Motivation and quick overview of the diffusion process
3. Attention-U-Net backbone: CNN/U-Net, time embedding, and cross-attention
4. Text conditioning: word vectors, transformers, and CLIP
5. Latent Diffusion Models: VAE compression and the Stable Diffusion pipeline
6. Applications: Stable Video Diffusion, GLIDE, DALL·E 2, Imagen, SD 2.0, SD 3.5, Nano Banana 2
7. Summary, limitations, and references

What is Stable Diffusion?

- ▶ A state-of-the-art text-to-image generative model.
- ▶ Released by Stability AI and collaborators (2022).
- ▶ Open-source, enabling wide adoption and customization.



Advantages

- ▶ High-quality, diverse image generation.
- ▶ Efficient training and inference due to latent space operations.
- ▶ Open-source and extensible for research and applications.

What do we need to make a good generative model?

1. Method of learning to generate new images
2. Way to link text and images
3. Way to compress images
4. Way to add in good inductive biases

Forward/reverse diffusion

Text-image representation model

Autoencoder

U-net architecture + 'attention'

Making a 'good' generative model is about making all these parts work together well!

Generative Modeling

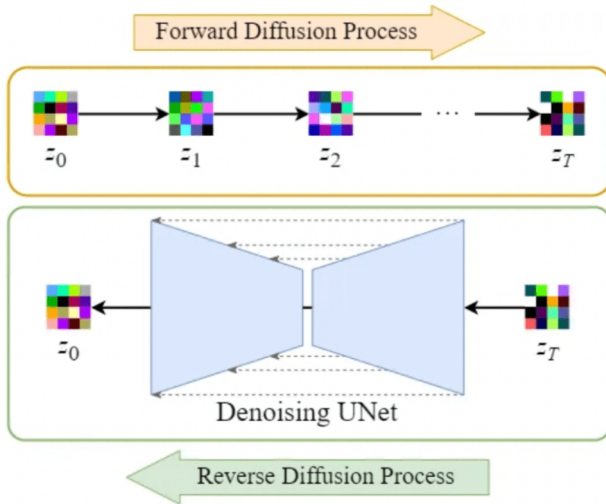
- ▶ Early successes: GANs and VAEs enabled impressive image synthesis and data generation.
- ▶ **Drawbacks:**
 - GANs: Training instability, mode collapse, and sensitivity to hyperparameters.
 - VAEs: Blurry outputs and limited expressiveness.
- ▶ **Key Challenge:** Achieving both high sample quality and stable, reliable training.

Where do we go from here?

- ▶ Need for models that are robust, interpretable, and capable of generating diverse, high-fidelity samples.

Enter Diffusion Models:

- ▶ Inspired by thermodynamics, diffusion models gradually transform noise into data.
- ▶ Highlights:
 - Simple training objective
 - Stable optimization
 - State-of-the-art sample quality
 - Probabilistic and interpretable framework

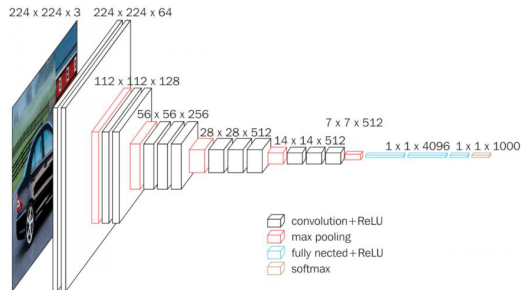


- ▶ **Forward:** Clean $\rightarrow$ Noise (additive Gaussian steps)
- ▶ **Reverse:** Learn **U-Net** to remove noise step by step
- ▶ **Training:** Model predicts the noise added
- ▶ **Sampling:** Roll back from pure noise to a data sample

Stable Diffusion Core Backbone:

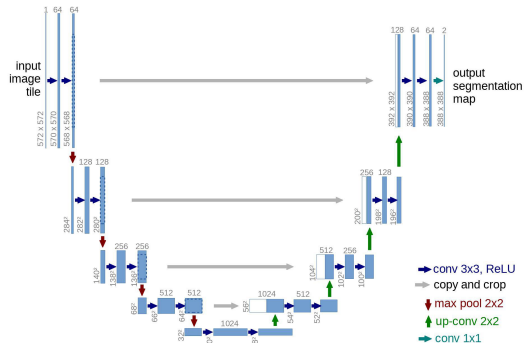
Attention-U-Net Architecture

- ▶ CNN parametrizes a function over images.
- ▶ **Motivation:**
 - Features are translationally invariant.
 - Extract features at different scales / abstraction levels.
- ▶ **Key modules:**
 - Convolution
 - Downsampling (max-pool)
- ▶ Features of larger scale (larger receptive field)
⇒ higher abstraction level.



- ▶ An inverted CNN (generator) can generate images.
- ▶ CNN + inverted CNN can model an image-to-image function.
- ▶ **Down sampling:** convolution.
- ▶ **Up sampling:** transposed convolution.

U-Net: A Natural Architecture for Image-to-Image Functions



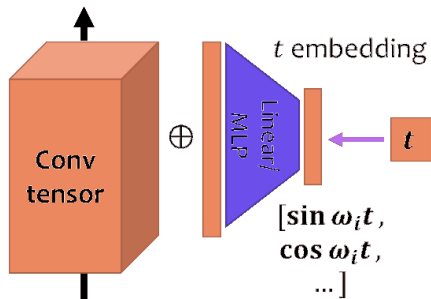
Note: Add Time Dependency

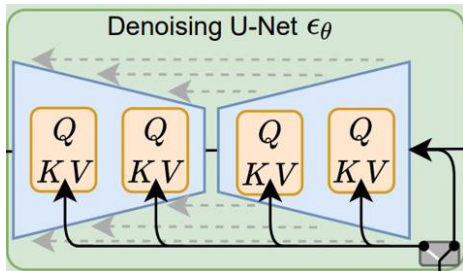
The score function is time-dependent.

$$\text{Target: } s(x, t) = \nabla_x \log p(x, t)$$

Add time dependency

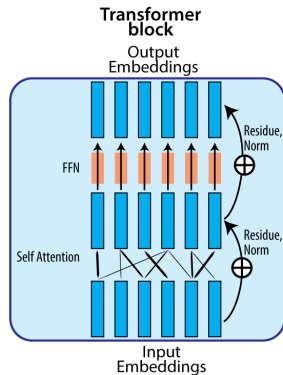
- ▶ Assume time dependency is spatially homogeneous.
- ▶ Add one scalar value per channel $f(t)$.
- ▶ Parametrize $f(t)$ by MLP/linear of Fourier basis.
 - Fourier features: $[\sin \omega_i t, \cos \omega_i t, \dots]$



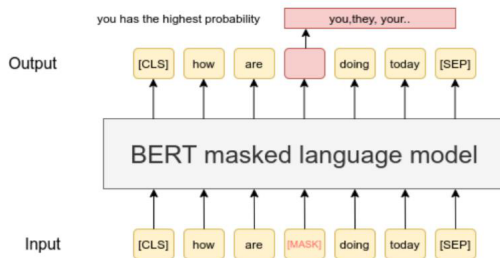


- ▶ Input convolution: Projects 4-channel input $\rightarrow$ 320 channels using a 2D convolution.
- ▶ Time embedding: inject time information into the network.
- ▶ Downsampling: sometimes interleaved with cross-attention.
- ▶ Mid block: applies additional residual and (cross-)attention operations.
- ▶ Upsampling, normalization, activation, output convolution.

- ▶ Meaning of a word depends on context, not always the same.
- ▶ Example: “I *book* a ticket to buy that *book*.”
- ▶ Word vectors should depend on context.
- ▶ Transformers let each word absorb influence from other words to become contextualized.

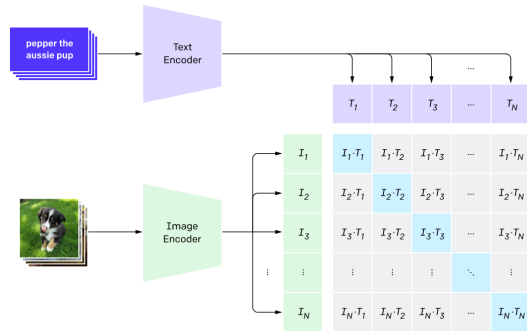


- ▶ Self-supervised learning of word representations.
- ▶ Predict missing / next words in a sentence (BERT, GPT).
- ▶ Contrastive learning: match image and text (CLIP).
- ▶ Downstream classifiers can decode part of speech, sentiment, ...



- ▶ Learn a joint encoding space for text captions and images.
- ▶ Maximize representation similarity between an image and its caption.
- ▶ Minimize similarity for other pairs.

1. Contrastive pre-training

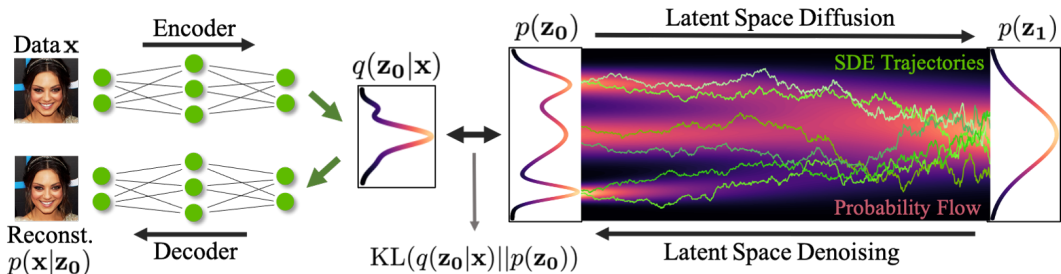


Adapted from: Radford et al., [CLIP](#), arXiv:2103.00020

- ▶ Encoder in Stable Diffusion: pre-trained CLIP ViT-L/14 text encoder.
- ▶ Word vectors can also be randomly initialized and learned online.
- ▶ **Other conditional signals:**
 - Object categories (e.g. Shark, Trout): one vector per class.
 - Face attributes (e.g. {female, blonde hair, with glasses}): one vector per attribute.
- ▶ Time to be creative!

Stable Diffusion: **A Latent Diffusion Model (LDM)**

What is a Latent Space?



- ▶ A **latent space** is an abstract, compressed space where high-dimensional data (like images) is represented in a lower-dimensional form.
- ▶ Think of it as a hidden layer of meaning—each point in this space represents a potential image or concept.
- ▶ In classical models like GANs or VAEs, the latent space is directly sampled from (e.g., a vector $\mathbf{z}$), and the generator turns it into an image.

What is a Latent Space? (cont.)

- ▶ **Latent** space = “Hidden space of ideas or features”

Two main ways latent spaces are used in diffusion:

► 1. Latent Diffusion Models (LDMs):

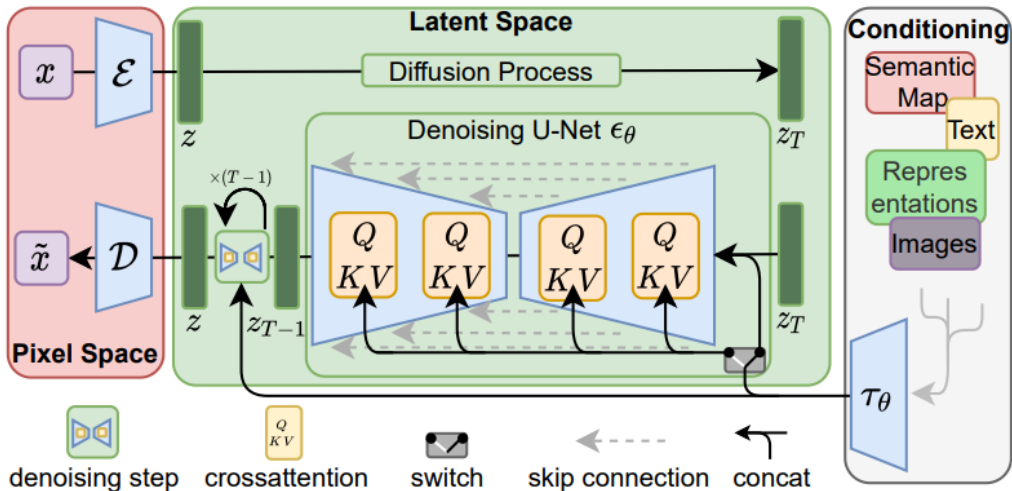
- Diffusion is performed in a compressed latent space, not directly on high-dimensional images.
- **Pipeline:** Compress image $\rightarrow$ Denoise in latent space $\rightarrow$ Decode back to image.
- Enables faster, cheaper training and inference, and allows for semantic control (e.g., text prompts, masks).

► 2. Latent Representations in Noise Vectors:

- Even in standard diffusion models, the initial noise vector acts as a latent code.
- Changing the noise vector generates different images or morphs between them.

- ▶ **Traditional pixel-space diffusion models are computationally expensive:** they operate directly on high-dimensional images, making both training and inference costly.
- ▶ **Latent diffusion reduces computational cost** by performing the diffusion process in a compressed, lower-dimensional latent space—while still maintaining high-quality image generation.
- ▶ This efficiency enables more complex and practical applications, such as text-guided generation and flexible image editing.
- ▶ **How is this achieved?** Instead of working in pixel space, we learn a compressed latent space using a Variational Autoencoder (VAE) with very low KL divergence weight (e.g., 10^{-6}). This yields a compact representation suitable for diffusion.
- ▶ This approach mirrors ideas from autoregressive models (e.g., VQGAN), enabling even more efficient and controllable generation pipelines.

Latent Diffusion Models: Visual Overview



Diffusion Models: **Applications**

- ▶ Initialize from Stable Diffusion.
- ▶ Add temporal layers (e.g., 1D Conv, temporal attention).
- ▶ Finetune on video data.
- ▶ Data pipeline:
 - Scrape **unlabeled** videos from the internet.
 - Use image and video captioners to generate synthetic text labels.
 - Apply aggressive data filtering using several heuristics (CLIP score, optical flow score, aesthetic scores, OCR, etc.).

Application: Stable Video Diffusion (cont.)





“a man wearing a white hat”

Image Inpainting with GLIDE

⁰GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models



a shiba inu wearing a beret and black turtleneck



a close up of a handpalm with leaves growing from it

Text to image generation with DALL.E 2



Fix the CLIP embedding z_i
Decode using different decoder latents $\mathbf{x}_T$

Image Variations



Interpolate image CLIP embeddings z_i

Use different x_T to get different interpolation trajectories.

Image interpolation



a photo of a cat → an anime drawing of a super saiyan cat, artstation



a photo of a victorian house → a photo of a modern house



a photo of an adult lion → a photo of lion cub

Change the image CLIP embedding towards the difference of the text CLIP embeddings of two prompts.

Decoder latent is kept as a constant.

Text Difference Image interpolation



A brain riding a rocketship heading towards the moon.



A dragon fruit wearing karate belt in the snow.



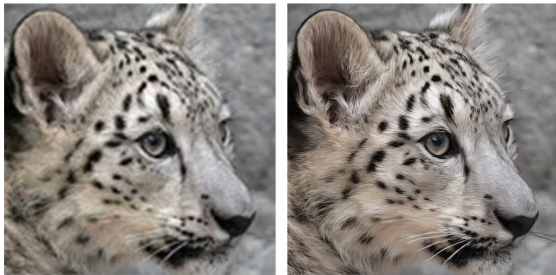
A relaxed garlic with a blindfold reading a newspaper while floating in a pool of tomato soup.

What changed vs. Stable Diffusion v1?

- ▶ **New text encoder:** OpenCLIP (LAION + Stability AI) replaces CLIP ViT-L/14—stronger language understanding and image quality.
- ▶ **Native resolutions:** text-to-image at **512×512** and **768×768**.
- ▶ **Upscaler diffusion:** ×4 super-resolution (e.g. 128→512).
- ▶ **depth2img:** depth-guided image-to-image—structure-preserving synthesis from inferred depth + text.
- ▶ **Updated inpainting:** text-guided inpainting fine-tuned on the Stable Diffusion 2.0 base model.

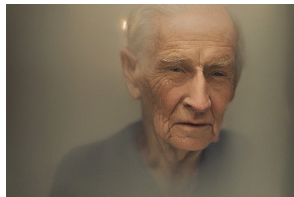
⁰Lopez, “Stable Diffusion 2.0 Release,” Stability AI (Nov 2022).

- ▶ Dedicated **upscaler diffusion** model enhances resolution by $\times 4$.
- ▶ Example: **128×128** low-res sample $\rightarrow$ **512×512** output.



Left: 128×128 . Right: 512×512 upscaled.

- ▶ **depth2img** infers depth from an input image, then generates new images conditioned on **text + depth**.
- ▶ Enables structure-preserving image-to-image and shape-conditional synthesis.
- ▶ Can produce radically different appearances while preserving spatial coherence and depth.



Input → variants (top); depth coherence preserved (bottom).



Text-guided inpainting model fine-tuned on the SD 2.0 text-to-image base.

⁰Stability AI, Stable Diffusion 2.0 Release.



Prompt: photo of three potions: the first potion is blue with the label "MANA", the second potion is red with the label "HEALTH", the third potion is green with the label "POISON". Old apothecary.

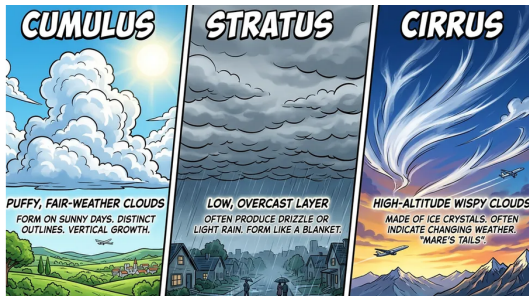
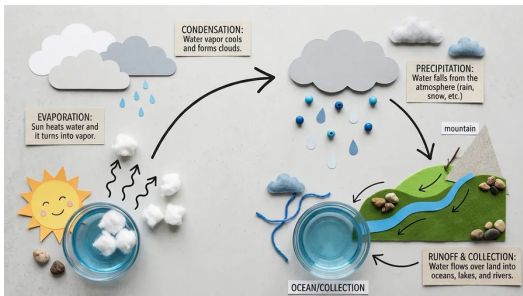


Prompt: scene of a giant ancient tortoise with a fantasy city built on its back. The tortoise's shell is covered in lush, dense forest with towering trees and a hidden, misty village nestled in the foliage. The city consists of intricately designed buildings that blend seamlessly with the natural environment, featuring rope bridges connecting different sections of the city.



Prompt: collage art "We're Leaving For the Future" 1980s #vaporwave aesthetic internet art glued layered magazine cutout image shape scrap, torn ragged paper art, BASIC code, halftone, #pixelart.

- **Customizability:** Large (8.1B), Large Turbo (8.1B) and Medium (2.5B) models.
- **Efficient Performance:** Optimized to run on standard consumer hardware; Medium and Large Turbo require no specialized GPUs.
- **Highly Accessible:** Medium needs only 9.9GB of VRAM making advanced diffusion imaging feasible on most consumer GPUs.



Examples: water-cycle infographic and cloud-type comparison.

- Builds on Google's Gemini Image line: Nano Banana (2025) → Nano Banana Pro → Nano Banana 2.
- Production-ready output: multiple aspect ratios and resolutions from 512px to 4K.

⁰Raisinghani, "Nano Banana 2: Combining Pro capabilities with lightning-fast speed," Google DeepMind blog (Feb 2026).

Text to Image generation with stable diffusion.
<https://huggingface.co/spaces/stabilityai/stable-diffusion>

Stable Diffusion: **Summary & Limitations**

From diffusion to Stable Diffusion

- ▶ Diffusion models learn to reverse a gradual noising process; sampling remains iterative but training is stable and scalable.
- ▶ **Latent Diffusion (LDM):** denoise in a VAE-compressed latent space ($\sim 8\times$ downsampling) rather than full pixel space—lower memory and faster training/inference.

Core architecture

- ▶ **Attention U-Net:** encoder–decoder with skip connections; downsampling / upsampling for multiscale features.
- ▶ **Time conditioning:** Fourier-feature time embeddings added per channel ($s(x, t) = \nabla_x \log p(x, t)$).
- ▶ **Text conditioning:** CLIP (ViT-L/14 in SD v1) maps prompts to token embeddings; cross-attention injects language into the U-Net.

Model evolution & applications

- ▶ **SD 2.0:** OpenCLIP text encoder; native 512/768 T2I; upscaler ($\times 4$), depth2img, and updated inpainting.
- ▶ **SD 3.5:** Large 8.1B, Large Turbo, Medium 2.5B; strong prompt adherence; run on consumer GPUs.
- ▶ Broader ecosystem: video (Stable Video Diffusion), inpainting/editing (GLIDE, DALL·E 2), and newer multimodal systems (e.g. Nano Banana 2).

- ▶ **Slow sampling** — many denoising steps (partially addressed by distilled models such as SD 3.5 Large Turbo).
- ▶ **Data & compute** — high-quality open models still require large-scale curated data and substantial GPU resources to train.
- ▶ **Prompt sensitivity** — output quality and adherence depend on prompt wording; base models may vary across seeds.
- ▶ **Distribution gaps** — rare concepts, fine details, and compositional reasoning remain challenging.
- ▶ **Safety & misuse** — text-to-image systems raise concerns around deepfakes, copyright, and misinformation.

Tutorials and Surveys

- ▶ Binxu Wang: [Stable Diffusion: A Tutorial \(Harvard ML from Scratch\)](#) — U-Net, time embedding, CLIP conditioning.
- ▶ Kreis, K., Gao, R., & Vahdat, A.: [CVPR 2022 Tutorial: Diffusion Models](#)
- ▶ Rogge, L., & Rasul, K.: [The Annotated Diffusion Model](#)
- ▶ Lilian Weng: [What are Diffusion Models?](#)
- ▶ Yang Song: [Score-Based Generative Modeling: A Brief Overview](#)

Foundational Diffusion Models

- ▶ Ho, J., Jain, A., & Abbeel, P. (2020). [Denoising Diffusion Probabilistic Models](#)
- ▶ Sohl-Dickstein, J., et al. (2015). [Deep Unsupervised Learning using Nonequilibrium Thermodynamics](#)
- ▶ Song, Y., & Ermon, S. (2019). [Generative Modeling by Estimating Gradients of the Data Distribution](#)
- ▶ Nichol, A., & Dhariwal, P. (2021). [Improved Denoising Diffusion Probabilistic Models](#)

Stable Diffusion: Core Papers

- ▶ Rombach, R., et al. (2022). **High-Resolution Image Synthesis with Latent Diffusion Models** — Stable Diffusion / LDM (VAE + U-Net + conditioning).
- ▶ Radford, A., et al. (2021). **Learning Transferable Visual Models From Natural Language Supervision (CLIP)** — text encoder and cross-attention conditioning.

Applications and Model Releases

- ▶ Nichol, A., et al. (2021). [GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models](#)
- ▶ Ramesh, A., et al. (2022). [Hierarchical Text-Conditional Image Generation with CLIP Latents \(DALL·E 2\)](#)
- ▶ Saharia, C., et al. (2022). [Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding \(Imagen\)](#)
- ▶ Blattmann, A., et al. (2023). [Stable Video Diffusion: Scaling Latent Video Diffusion Models to Large Datasets](#)
- ▶ Lopez, J. (2022). [Stable Diffusion 2.0 Release](#) — OpenCLIP, upscaler, depth2img, inpainting (Stability AI blog).
- ▶ Stability AI (2024). [Introducing Stable Diffusion 3.5](#) — SD 3.5 Large / Large Turbo / Medium (Stability AI blog).

- Raisinghani, A. (2026). [Nano Banana 2: Combining Pro capabilities with lightning-fast speed](#) (Google DeepMind blog).